Beginning MATLAB and Simulink

From Novice to Professional

Sulaymon Eshkabilov

Apress®

Beginning MATLAB and Simulink: From Novice to Professional

Sulaymon Eshkabilov
Fargo, United States

ISBN-13 (pbk): 978-1-4842-5060-0 ISBN-13 (electronic): 978-1-4842-5061-7
https://doi.org/10.1007/978-1-4842-5061-7

Managing Director, Apress Media LLC: Welmoed Spahr
Acquisitions Editor: Steve Anglin
Development Editor: Matthew Moodie
Coordinating Editor: Mark Powers

Cover designed by eStudioCalamar

Cover image from rawpixel.com

Distributed to the book trade worldwide by Springer Science+Business Media New York, 233 Spring Street, 6th Floor, New York, NY 10013. Phone 1-800-SPRINGER, fax (201) 348-4505, e-mail orders-ny@springer-sbm.com, or visit www.springeronline.com. Apress Media, LLC is a California LLC and the sole member (owner) is Springer Science + Business Media Finance Inc (SSBM Finance Inc). SSBM Finance Inc is a **Delaware** corporation.

For information on translations, please e-mail editorial@apress.com; for reprint, paperback, or audio rights, please email bookpermissions@springernature.com.

Apress titles may be purchased in bulk for academic, corporate, or promotional use. eBook versions and licenses are also available for most titles. For more information, reference our Print and eBook Bulk Sales web page at http://www.apress.com/bulk-sales.

Any source code or other supplementary material referenced by the author in this book is available to readers on GitHub via the book's product page, located at www.apress.com/9781484250600. For more detailed information, please visit http://www.apress.com/source-code.

Printed on acid-free paper

To the memory of my father.
To my mother.
To my wife, Nigora, after 25 wonderful years together.

Table of Contents

About the Author

Sulaymon Eshkabilov, PhD, is currently a visiting professor at the Department of Agriculture and Biosystems, North Dakota State University, USA. He obtained an ME diploma from the Tashkent Automobile Road Institute, an MSc from Rochester Institute of Technology, NY, USA, and a PhD from Cybernetics Institute of Academy Sciences of Uzbekistan in 1994, 2001, and 2005, respectively. He was an associate professor at the Tashkent Automobile Road Institute from December 2006 until January 2017. He held visiting professor and researcher positions at Ohio University, USA, during the 2010/2011 academic year and at Johannes Kepler University, Austria from January-September, 2017. He teaches the following courses: "MATLAB/Simulink Applications for Mechanical Engineering and Numerical Analysis" and "Modeling of Engineering Systems" to undergraduate students, "Advanced MATLAB/Simulink Modeling" seminar/class, "Control Applications," "System Identification," and "Experimentation and Testing with Analog and Digital Devices" to graduate students.

His research interests are mechanical vibrations, control, mechatronics, and system dynamics. He developed simulation and data analysis models for vibrating systems, autonomous vehicle control, and studies of mechanical properties of bones. He has authored two books devoted to MATLAB/Simulink applications for mechanical engineering students and numerical analysis. He has worked as an external academic expert in the European Commission to assess academic projects in 2019.

About the Technical Reviewers

Dr. Joseph Mueller specializes in control systems and trajectory optimization. For his doctoral thesis, he developed optimal ascent trajectories for stratospheric airships. His active research interests include robust optimal control, adaptive control, applied optimization and planning for decision support systems, and intelligent systems to enable autonomous operations of robotic vehicles. Prior to joining SIFT in early 2014, Dr. Mueller worked at Princeton Satellite Systems for 13 years. In that time, he served as the principal investigator for eight small business innovative research contracts for NASA, the Air Force, the Navy, and MDA. He has developed algorithms for optimal guidance and control of both formation flying spacecraft and high altitude airships and developed a course of action planning tool for DoD communication satellites. In support of a research study for the NASA Goddard Space Flight Center in 2005, Dr. Mueller developed the Formation Flying Toolbox for MATLAB, a commercial product that is now used at NASA, ESA, and several universities and aerospace companies around the world. In 2006, Dr. Mueller developed the safe orbit guidance mode algorithms and software for the Swedish Prisma mission, which has successfully flown a two-spacecraft formation flying mission since its launch in 2010. Dr. Mueller also serves as an adjunct professor in the Aerospace Engineering & Mechanics Department at the University of Minnesota, Twin Cities campus.

Karpur Shukla is a research fellow at the Centre for Mathematical Modeling at FLAME University in Pune, India. His current research interests focus on topological quantum computation, nonequilibrium, and finite-temperature aspects of topological quantum field theories, and applications of quantum materials effects for reversible computing. He received an M.Sc. in physics from Carnegie Mellon University, with a background in theoretical analysis of materials for spintronics applications as well as Monte Carlo simulations for the renormalization group of finite-temperature spin lattice systems.

Acknowledgments

I would like to express my special gratitude to the technical reviewers of this book—Dr. Joseph Mueller and Mr. Karpur Shukla—for their very thorough work while reviewing the context and code of the book. Without their critical insights and corrections many points along the way, the book would not be as good as it is now. I would also like to express my special gratitude to Mark Powers for his timely and well planned correspondence throughout this book project.

My cordial gratitude goes to my mother, for her limitless support and love. Up until the last point of this book, she has kept checking my progress.

I would like to thank my wife, Nigora. Without her great support, I would not be able to take up the challenging task of writing this book. I have spent so much time over the weekends in my office writing and editing the book. In addition, I would like to thank our children—Anbara, Durdona, and Dovud—for being such delightful people and for being the inspiration for writing this book.

Introduction

This book is aimed at beginner-level learners of MATLAB/Simulink packages. It covers the essential, hands-on tools and functions of the MATLAB and Simulink packages, and explains them via interactive examples and case studies. The main principle of the book is "learning by doing" and it progresses from simple to complex. It contains dozens of solutions and simulation models via M/MLX files/scripts and Simulink models, which help you learn the programming and modeling essentials. Moreover, there are many recommendations for avoiding pitfalls related to programming and modeling aspects of MATLAB/Simulink packages.

"Beginning MATLAB and Simulink" explains various practical issues of programming and modeling in parallel by comparing programming tools of MATLAB and blocks of Simulink. After studying this book, you'll be proficient at using MATLAB/Simulink packages. You can apply the source code and models from the book's examples as templates to your own projects in data science, numerical analysis, modeling and simulation, or engineering.

Essential learning outcomes of the book include:

- Getting started using MATLAB and Simulink
- Performing data analysis and visualization with MATLAB
- Programming essentials of MATLAB and core modeling aspects of Simulink, and how to associate scripts and models of MATLAB and Simulink packages
- Developing GUI models and standalone applications in MATLAB
- Working with integration and numerical root-finding methods in MATLAB and Simulink
- Solving differential equations in MATLAB and Simulink
- Applying MATLAB for data analysis and data science projects

The book contains eight logically interlinked chapters.

- Chapter 1 is dedicated to introducing the MATLAB environment and MATLAB recognized data types, including numeric, cell, structure, character, logical, function handle, and table.
- Chapter 2 covers most of the essential programming tools and functions, such as `for … end` and `while … end` loop operators, `if…elseif…else…end` condition operators, symbol referencing, and most common errors and warnings. It also covers M-file debugging tools and options in MATLAB. Moreover, this chapter addresses MATLAB-specific programming tools, including function files, function handles, and display operators, such as `disp()`, `display`, `fprintf`, and `sprint`.
- Chapter 3 covers GUI development and how to write and edit GUI model callback functions.
- Chapter 4 addresses the issues of how to develop MEX files, C/C++ code, and standalone applications from the existing M-files.
- Chapter 5 is dedicated to Simulink modeling essentials. The examples in this chapter cover matrix operations, computing function values, modeling, and solving ordinary differential equations. Moreover, it covers issues around how to associate Simulink models with MATLAB scripts.
- Chapter 6 is devoted to data visualization issues, such as building 2D and 3D plots and animated plots in MATLAB.
- Chapter 7 is dedicated to matrix algebra and array operations. It addresses solving systems linear equations, eigen-value problems, matrix decompositions, matrix and vector operations, and conversions of arrays and strings via examples solved in MATLAB and Simulink in parallel.
- Chapter 8 covers some essential aspects of solving ordinary differential equations analytically and using MATLAB's built in functions and commands.

All of the source code (scripts, M/MLX/MAT files, Simulink models, SLX/MDL files, C code, MEX-files, and installation `*.exe` files) discussed in the book are available to readers via a Download Source Code button on the book's `apress.com` product page and accessible via the Download Source Code link located at www.apress.com/9781484250600.

Note The scripts in the book may not always be the best solutions to the given problems, but this is done intentionally in order to emphasize methods used to improve them. In some other cases, it is found to be the most appropriate solution to the author's best knowledge. Should I spot better alternative solutions to exercises, I will publish them via MathWorks' MATLAB Central User Community's file exchange, via my file exchange link there (under my name).

No matter how hard we worked to proofread this book, it is inevitable that there will be some typographical errors that might slip through and will appear in print.

September 2019 Sulaymon Eshkabilov